IMPACT ANALYSIS CR-001: Advanced Medicine Search and Filtering

1. Change Summary

CR-001 adds advanced medicine search and combined filters for medicine name, category, supplier, price range, and stock availability. Users can clear filters and view the number of matching medicines.

2. Impact Assessment

Area	Impact	Assessment
Frontend	Moderate	Add controls, combined filtering logic, result count, and empty state.
Backend	None–Low	Reuse existing medicine data; changes only if required fields are missing.
API	None–Low	Existing endpoints expected to remain in use.
Database	None expected	No schema migration is planned.
Authentication	None	Existing login/session handling unchanged.
Authorization	None	Existing role permissions must remain enforced.
Performance	Low–Moderate	Client-side filtering depends on number of loaded medicines.
CRUD	Low	Existing actions must retain original medicine IDs.
Deployment	Low	Frontend asset update; possible API adjustment only if necessary.

3. Frontend Impact

The main affected area is the Medicine Management page.

- Add name search, category and supplier selectors, price inputs, and stock-status selector.
- Apply all filters together to the medicine records.
- Display matching count, Clear All Filters, and no-result message.
- Keep controls responsive and preserve edit/delete actions on filtered rows.

4. Backend, API and Database Impact

The preferred approach is client-side filtering of the existing medicine list. No new endpoint or database schema change is planned.

Before implementation, confirm that the medicine response includes category, supplier, price, quantity, and the threshold needed to identify low stock. If a required field is absent, a limited existing API-response adjustment may be necessary and must be documented.

Do not alter stock or sales records as part of filtering.

5. Security Impact

Existing authentication and role-based authorization must remain unchanged.

- Do not use filtering to bypass protected medicine APIs.
- Do not expose additional data beyond the current user role.
- Treat search and price inputs as untrusted values; use safe DOM text rendering.
- Ensure frontend filtering does not replace backend authorization checks.

6. CRUD and Functional Impact

Filtering changes only which medicine rows are displayed. It must not modify inventory quantities or medicine identities.

- Edit and Delete must continue to target the original medicine ID.
- After create/update/delete, the displayed results and count should refresh consistently.
- Stock-status classification must align with the system’s existing low-stock threshold.
- Handle missing supplier, price, or quantity fields without crashing.

7. Performance Impact

Client-side filtering is expected to be responsive for the current medicine list size. Avoid an API call on every keystroke when all required data is already loaded.

For a very large inventory, server-side filtering or pagination may be a future enhancement. Actual performance should be checked with representative data.

8. Testing Impact

- Medicine name: exact, partial, mixed case, and whitespace.
- Category and supplier selection.
- Minimum/maximum price: boundaries, empty fields, invalid range, nonnumeric input.
- Available, Low Stock, and Out of Stock classification.
- Multiple simultaneous filters using AND logic.
- Result count, no-results message, and Clear All Filters.
- Missing/null data and responsive layout.
- Edit/Delete on filtered rows and existing access control.

9. Risk Analysis

Risk	Impact	Mitigation
Incorrect stock status	High	Confirm and test the system’s reorder-level rule.
Wrong medicine edited/deleted	High	Retain original medicine IDs; regression test actions.
Missing API fields	Medium	Inspect response and document any minimal API change.
Incorrect combined filtering	Medium	Test AND logic and boundary cases.
Slow filtering on large lists	Medium	Measure responsiveness; consider pagination later.

Risk	Impact	Mitigation
Poor mobile layout	Low	Test small and large screen widths.

10. Overall Impact and Recommendation

Overall Impact: Low–Moderate

CR-001 is a feasible usability enhancement focused on the Medicine Management frontend. It is recommended for implementation after confirming the existing medicine data fields and stock-status rules, followed by functional, security, and regression testing.

Key consideration: Filtering must not alter medicine IDs, stock values, or existing access permissions.